

Classical Mechanics

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Preface

This is my notes on Classical Mechanics. The main reference is *Mechanics* by L.D. Landau and E.M. Lifshitz and Other references include *Classical Mechanics* by Herbert Goldstein, Charles Poole, and John Safko,

Introduction to Mechanics and Symmetry: A Basic Exposition to Classical Mechanical Systems by Jerrold E. Marsden and Tudor S. Ratiu, and *Foundations of Mechanics* by Ralph Abraham and Jerrold E. Marsden.

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Chapter 1

Classical Mechanics

1.1 Basic Concepts

1.2 Newton Fomalism

Axiom 1.2.1. 质点的状态需要用 4 个自由度 (t, x, y, z) 描述。质点的所有可能的运动状态构成 $\mathbb{R}^4$ 。

Definition 1.2.2. 称坐标系 Σ 为惯性系，若在 Σ 中孤立质点保持静止或匀速直线运动。

Axiom 1.2.3. 惯性系存在。

Theorem 1.2.4. 两个惯性系 $\Sigma = (t, x, y, z)$, $\Sigma' = (t', x', y', z')$ 之间的变换满足

$$\begin{pmatrix} t' \\ x' \\ y' \\ z' \end{pmatrix} = A \begin{pmatrix} t \\ x \\ y \\ z \end{pmatrix} + b \quad (1.1)$$

其中 $A \in M_4(\mathbb{R}), b \in \mathbb{R}^4$ 。

Axiom 1.2.5. 两个惯性系 $\Sigma = (t, x, y, z)$, $\Sigma' = (t', x', y', z')$ 之间的变换满足 $(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2 = (x'_1 - x'_2)^2 + (y'_1 - y'_2)^2 + (z'_1 - z'_2)^2$

Axiom 1.2.6. 惯性系中，在固定质点和其他物体的相互作用的情况下，质点的加速度和质量成反比。

Definition 1.2.7. 称 $\vec{F} = m\vec{a}$ 为质点受到的力。

Chapter 2

Lagrange Formalism

2.1 Generalized Coordinates

2.2 Lagrange Equation

2.3 Hamilton Principle

2.4 Integral of Motion

2.5 Symmetry

Chapter 3

Hamilton Formalism

3.1 Hamilton Canonical Equations

3.2 Poisson Bracket

3.3 Phase Space

3.4 Canonical Transformation

3.5 Hamilton-Jacobi Equation

3.6 Angle-Action Variables

Chapter 4

Exact Solutions

4.1 Central Force Field

4.2 Harmonic Oscillator

4.3 Rigid Body

Chapter 5

Stability and Chaos

Chapter 6

Introduction to Symplectic Geometry

Chapter 7

Special Relativity

7.1 Velocity of Light and Locality

7.2 Axiomatic System of Special Relativity

Axiom 7.2.1. 质点的状态需要用 4 个自由度 (t, x, y, z) 描述。质点的所有可能的运动状态构成 $\mathbb{R}^4$ 。

Definition 7.2.2. 称坐标系 Σ 为惯性系，若在 Σ 中孤立质点保持静止或匀速直线运动。

Axiom 7.2.3. 惯性系存在。

Theorem 7.2.4. 两个参考系 $\Sigma = (t, x, y, z)$, $\Sigma' = (t', x', y', z')$ 之间的变换满足

$$\begin{pmatrix} t' \\ x' \\ y' \\ z' \end{pmatrix} = A \begin{pmatrix} t \\ x \\ y \\ z \end{pmatrix} + b \quad (7.1)$$

其中 $A \in M_4(\mathbb{R}), b \in \mathbb{R}^4$ 。

Axiom 7.2.5. 在任意参考系中，质点的速度存在上限 c 。

Axiom 7.2.6. 惯性系中，在固定质点和其他物体的相互作用的情况下，质点的加速度和质量成反比。

Definition 7.2.7. 称 $\vec{F} = m\vec{a}$ 为质点受到的力。

7.3 Electrodynamics in Special Relativity